

Before you can access the Graphics2D capabilities, you must perform a downcast on the Graphics g reference passed to paint. This is safe because—trust me—the object is actually of type Graphics2D:

```
public void paint(Graphics g )
{
    Graphics2D g2d = ( Graphics2D ) g;
    Class Graphics versus Graphics2D
```

Normally, we would hesitate to cast a Superclass reference to a Subclass type, because we know the Subclass is likely to have many methods not present in the Superclass. However, this is safe merely because we know g is really a Subclass object. Anytime you wish to convert a Graphics reference into a Graphics2D reference, you must use this downcast.

```
public void paint(Graphics g )
{
    Graphics2D g2d = ( Graphics2D ) g; 
```